The smallest I-contractible graphs that are not strong I-contractible, and odd torsion in flag complexes on thirteen vertices

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Abstract

Ivashchenko's contractible transformations define a hierarchy of graph classes $\mathfrak{I}_{VS} \subseteq \mathfrak{I}_S \subseteq \mathfrak{I}$ (vertex-strong, strong and plain I-contractible graphs) whose clique complexes are contractible. Dochtermann, Espinoza, Frías-Armenta and Hernández [Discrete Math. 346 (2023) 113516] exhibited a 15-vertex flag triangulation of the dunce hat lying in $\mathfrak{I} \setminus \mathfrak{I}_S$ and conjectured that it is the smallest graph with trivial $\mathbb{Z}_2$-homology that is not strong I-contractible, and in particular the smallest member of $\mathfrak{I} \setminus \mathfrak{I}_S$. We refute both statements by an exhaustive, certified computer search of 2.1×10^{13} graphs on 13 vertices and 6.5×10^{13} K_4-free graphs on 14 vertices, carried out on the YUCA cluster after a reduction to "stuck" graphs (no vertex and no edge admits a strong contractible deletion). The smallest $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$ have 13 vertices: there are exactly three, and the clique complex of each has $H_1 \cong \mathbb{Z}/3$, so they are Moore spaces $M(\mathbb{Z}/3, 1)$ and—by a result of Adamaszek on torsion in flag complexes with at most 12 vertices—the smallest flag complexes with odd torsion. The smallest members of $\mathfrak{I} \setminus \mathfrak{I}_S$ have 14 vertices; within the K_4-free class there are exactly three stuck ones, each a flag triangulation of a disc whose boundary is identified along a cycle of length 4, 5 or 6 with the dunce-hat pattern $a\,a\,a^{-1}$. The same examples refute Conjecture 15 of the cited paper with exact minimum 14. We extend the enumeration of the classes $\mathfrak{I}$, $\mathfrak{I}_S$, $\mathfrak{I}_{VS}$ and k-dismantlable graphs to 12 vertices, give pseudocode for all algorithms, and release code, data and machine-checkable certificates.

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1. Introduction

In a series of papers motivated by molecular spaces and digital topology, Ivashchenko [1, 2, 3] introduced four *contractible transformations* of a finite simple graph: deleting or adding a vertex whose open neighbourhood is itself contractible, and deleting or adding an edge uv whose common neighbourhood $N(u) \cap N(v)$ is contractible. The graphs that can be reduced to a single vertex by such transformations form the class $\mathfrak{I}$ of *I-contractible* graphs. Chen, Yau and Yeh [4] proved that contractible transformations preserve the simple homotopy type of the clique complex $\Delta(G)$, so every graph in $\mathfrak{I}$ has a contractible clique complex; very recently Dochtermann and Matsushita [5] proved the converse, so that $G \in \mathfrak{I}$ if and only if $\Delta(G)$ is contractible, and that I-homotopy of graphs coincides with simple homotopy equivalence of clique complexes. Contractible transformations have since been connected to strong homotopy types [6], to simplicial simple-homotopy of flag complexes [7, 8], to dismantlability hierarchies [9], and to collapsibility questions for flag complexes [10, 11].

Because the definition of $\mathfrak{I}$ allows edges and vertices to be *added*, deciding membership is delicate. Two natural subclasses are easier to handle [12]: the *strong I-contractible* graphs $\mathfrak{I}_S$, which can be reduced to a vertex using deletions only (of vertices with $N(v) \in \mathfrak{I}_S$ and edges with $N(u, v) \in \mathfrak{I}_S$), and the *vertex-strong I-contractible* graphs $\mathfrak{I}_{VS}$, which can be reduced using vertex deletions only. One has

$$\mathfrak{I}_{VS} \subseteq \mathfrak{I}_S \subseteq \mathfrak{I} \subseteq \{G : \Delta(G) \text{ is contractible}\},$$

with the last inclusion an equality by [5]. This is the graph-theoretic shadow of the classical chain *non-evasive* $\subset$ *collapsible* $\subset$ *contractible* for simplicial complexes [13, 14], and, as for complexes, the interesting questions concern the gaps. Dochtermann, Espinoza, Frías-Armenta and Hernández [12] determined by computer the smallest members of $\mathfrak{I}_S \setminus \mathfrak{I}_{VS}$ (eleven vertices), computed the cardinalities of all these classes for connected graphs on at most eleven vertices, and exhibited a 15-vertex flag triangulation D_{15} of Zeeman's

dunce hat [15] which is I-contractible but not strong I-contractible (Fig. 2A of [12]). Since $\mathfrak{I}_S$ is contained in the class of graphs whose clique complex is collapsible [10], and the dunce hat is the standard example of a contractible non-collapsible complex, this is a natural candidate for a smallest example, and the authors conjectured:

> **Conjecture 14 of [12].** *The smallest graph (in terms of vertices) with trivial $\mathbb{Z}_2$-homology that is not strong I-contractible is shown in Fig. 2A. In particular this is the smallest graph that is I-contractible but not strong I-contractible.*

A companion statement, Conjecture 15 of [12], asserts that the graph of Fig. 2B (the dunce hat with one extra edge) is the smallest strong I-contractible graph G having an edge $\{v, w\}$ with $N(v, w) \in \mathfrak{I}_S$ such that $G \setminus \{v, w\}$ is not strong I-contractible—that is, the smallest graph in which a "legal" strong deletion of an edge destroys strong I-contractibility.

In this paper we settle both conjectures in the negative by an exhaustive and independently certified computer search, and we find that the two sentences of Conjecture 14 fail for *different* reasons.

Main results. Let $\Delta(G)$ denote the clique complex. A graph is $\mathbb{Z}_2$-*acyclic* if $\widetilde{H}_*(\Delta(G); \mathbb{Z}_2) = 0$.

(A) (*Thirteen vertices, torsion*; Theorem 6.1.) Every $\mathbb{Z}_2$-acyclic graph on at most 12 vertices is strong I-contractible. On 13 vertices there are exactly three $\mathbb{Z}_2$-acyclic graphs that are not strong I-contractible. Their clique complexes have $H_1(\,\cdot\,; \mathbb{Z}) \cong \mathbb{Z}/3$ and $H_2 = 0$: they are 2-dimensional Moore spaces $M(\mathbb{Z}/3, 1)$, hence not contractible and in particular not I-contractible. The first sentence of Conjecture 14 is therefore false, but not because of a smaller dunce hat: it is false because $\mathbb{Z}_2$-homology does not see odd torsion. By Adamaszek's classification of torsion in flag complexes with at most 12 vertices [16], these are the smallest flag complexes with odd torsion in H_1.

(B) (*Fourteen vertices, the gap* $\mathfrak{I} \setminus \mathfrak{I}_S$; Theorem 6.4.) No graph on at most 13 vertices is I-contractible but not strong I-contractible, and there are such graphs on 14 vertices. Among K_4-free graphs on 14 vertices (equivalently, graphs whose clique complex is 2-dimensional, the class containing the dunce hat) there are exactly 51 *stuck* $\mathbb{Z}_2$-acyclic graphs

that are not in $\mathfrak{I}_S$: three of them, $G^a_{14}, G^b_{14}, G^c_{14}$, are contractible and I-contractible, and the remaining 48 have $H_1 \cong \mathbb{Z}/3$. The second sentence of Conjecture 14 is therefore false, with exact minimum 14 instead of 15.

(C) (*Reduction*; Theorem 3.7.) A smallest $\mathbb{Z}_2$-acyclic graph outside $\mathfrak{I}_S$ with the minimum number of edges is *stuck*: no open neighbourhood $N(v)$ and no non-empty common neighbourhood $N(u, v)$ is $\mathbb{Z}_2$-acyclic. Stuck graphs on at most 22 vertices are 2-connected, have minimum degree at least 4 and no vertex of degree $n - 1$. This reduces the search space by three to four orders of magnitude and is what makes the enumeration feasible.

(D) (*Conjecture 15*; Theorem 7.1.) The smallest strong I-contractible graph with an edge $\{v, w\}$, $N(v, w) \in \mathfrak{I}_S$, whose deletion leaves a graph outside $\mathfrak{I}_S$ has 14 vertices: $G^a_{14} + \{0, 1\}$ is such a graph, and there is none on 13 vertices.

(E) (*Table*; Section 5.) We recompute the cardinality table of [12] for $n \leq 11$, obtaining identical values in all cells, add the column $\mathfrak{I}$, and extend the table to $n = 12$ (connected graphs: 164,059,830,476) with the new value $|\mathfrak{I}_{VS}(12)| = 22{,}666{,}525{,}713$; rows for $n = 13$ and for K_4-free graphs on 14 vertices are in progress and reported where available.

Methodological remark. The examples of (A) show that $\mathbb{Z}_2$-homology, which is the natural invariant to compute in a search of this size, is not sufficient in this problem: the integer homology of candidates must be computed, and the finite certificates we provide (deletion sequences, Smith normal forms) are over $\mathbb{Z}$. Our examples in (B) are, like D_{15}, flag triangulations of a disc with its boundary identified along a short cycle with the dunce-hat pattern $a\,a\,a^{-1}$; the 48 torsion examples realise the pattern a^3 on cycles of length 4, 5 and 6. We conjecture that this picture persists (Section 7.3).

Reproducibility. All code, the exact cluster scripts, the raw logs, every graph found and human-readable certificates are available in a public repository,

`https://github.com/Geometric-and-Combinatorial-Structures/minimal-I-contractible-not-strong`

A script `verify.sh` re-checks every finite certificate behind Theorems 6.1, 6.4 and 7.1 with a dependency-free Python implementation, independent of the C code used in the search, in under a minute on a laptop.

Organisation. Section 2 recalls the definitions. Section 3 proves the reduction to stuck graphs. Section 4 describes the algorithms, the distributed computation and its validation. Section 5 presents the extended table. Section 6 describes the minimal examples in detail. Section 7 discusses consequences (including the conjectures of [17]) and open problems. The appendices contain the proofs of the reduction lemmas, the list of the 51 stuck graphs, the complete certificates and the cluster accounting.

2. Preliminaries

All graphs are finite and simple. For a graph G and a vertex v we write $N(v) = N_G(v)$ for the *open* neighbourhood (the subgraph induced by the vertices adjacent to v), and for an edge uv we write $N(u, v) = N(u) \cap N(v)$ for the *common neighbourhood*, again as an induced subgraph. $G - v$ and $G - uv$ denote deletion of a vertex and of an edge; $G + uv$ denotes the addition of an edge between non-adjacent vertices. The *clique complex* $\Delta(G)$ is the simplicial complex whose simplices are the cliques of G; a complex of this form is called *flag*. Note that $\Delta(N(v))$ is the link of v in $\Delta(G)$ and $\Delta(N(u, v))$ is the link of the edge uv.

2.1. Contractible transformations

We follow the conventions of [12].

Definition 2.1 ([2])**.** The class $\mathfrak{I}$ of *I-contractible* graphs is the smallest class of graphs containing K_1 and closed under the following four transformations:

(I1) deleting a vertex v with $N(v) \in \mathfrak{I}$;

(I2) adding a new vertex v adjacent to the vertices of an induced subgraph $H \in \mathfrak{I}$;

(I3) deleting an edge uv with $N(u, v) \in \mathfrak{I}$;

(I4) adding an edge uv between non-adjacent vertices with $N(u, v) \in \mathfrak{I}$.

The four transformations are the *contractible transformations* of Ivashchenko; (I3) and (I4) can be realised by (I1) and (I2) [4, Lemma 3]. Since contractible transformations do not change the simple homotopy type of the clique complex [4], every graph in $\mathfrak{I}$ has contractible $\Delta(G)$; by [5] the converse also holds.

Definition 2.2 ([12])**.** The class $\mathfrak{I}_S$ of *strong I-contractible* graphs is the smallest class containing K_1 such that, whenever $G \in \mathfrak{I}_S$,

(S1) $G - v \in \mathfrak{I}_S$ for every vertex v with $N(v) \in \mathfrak{I}_S$, and

(S2) $G - uv \in \mathfrak{I}_S$ for every edge uv with $N(u, v) \in \mathfrak{I}_S$.

Equivalently, $G \in \mathfrak{I}_S$ if and only if G can be reduced to K_1 by a sequence of deletions of vertices v with $N(v) \in \mathfrak{I}_S$ and edges uv with $N(u, v) \in \mathfrak{I}_S$. The class $\mathfrak{I}_{VS}$ of *vertex-strong I-contractible* graphs is defined in the same way using only (S1).

The empty graph is *not* in $\mathfrak{I}_S$; in particular an edge uv with $N(u, v) = \emptyset$ is never deleted, and an isolated vertex is never deleted. A vertex or edge satisfying the condition in (S1) or (S2) will be called $\mathfrak{I}_S$*-deletable*. In a graph of $\mathfrak{I}_S$ the order of deletions matters: a sequence may lead to a graph with no $\mathfrak{I}_S$-deletable element while another sequence reaches K_1 ([12, Fig. 3]); thus a greedy procedure does not decide $\mathfrak{I}_S$ and an exact search is needed (Section 4).

Definition 2.3 ([12, 9])**.** A vertex v of G is 0*-dismantlable* if $N(v)$ is a cone (some vertex of $N(v)$ is adjacent to all others), and, for $k \geq 1$, k*-dismantlable* if $N(v)$ is $(k-1)$-dismantlable; (-1)-dismantlable graphs are the cones. A graph is k*-dismantlable* (k-Dis) if it can be reduced to K_1 by deleting k-dismantlable vertices. A 0-dismantlable graph is also called *dismantlable* or *cop-win* [18].

One has (-1)-Dis $\subseteq$ 0-Dis $\subseteq$ 1-Dis $\subseteq \cdots \subseteq \mathfrak{I}_{VS} \subseteq \mathfrak{I}_S \subseteq \mathfrak{I}$ ([12, Lemma 19] and induction).

2.2. Homology

We write $\widetilde{H}_i(G; R)$ for the reduced simplicial homology of $\Delta(G)$ with coefficients in a ring R, and $\chi(G)$ for the Euler characteristic of $\Delta(G)$, which can be computed from the clique numbers $c_k(G)$ as $\chi(G) = \sum_{k \geq 1} (-1)^{k-1} c_k(G)$. A graph is $\mathbb{Z}_2$*-acyclic* if $\widetilde{H}_*(G; \mathbb{Z}_2) = 0$ (this includes connectedness), and $\mathbb{Z}$*-acyclic* if $\widetilde{H}_*(G; \mathbb{Z}) = 0$. By the universal coefficient theorem $\mathbb{Z}$-acyclic implies $\mathbb{Z}_2$-acyclic, and the two notions differ exactly when $H_*(G; \mathbb{Z})$ consists of odd torsion only; the first examples of this phenomenon among flag complexes are described in Section 6.1. A simplicial complex is *contractible* if it is homotopy equivalent to a point; for finite complexes contractible implies

$\mathbb{Z}$-acyclic, and the converse fails (the standard example being the dunce hat with a 2-cell attached differently, or acyclic 2-complexes with non-trivial perfect fundamental group), but for all the small complexes in this paper the two notions coincide as will be verified directly.

Integer homology of the small complexes in this paper is computed by exact Smith normal form of the boundary matrices (Section 4). We use the following elementary fact repeatedly.

Lemma 2.4. *Let n be the smallest order of a $\mathbb{Z}_2$-acyclic graph that is not in $\mathfrak{I}_S$ (if such graphs exist). Then for every graph H with fewer than n vertices, $H \in \mathfrak{I}_S$ if and only if H is $\mathbb{Z}_2$-acyclic.*

Proof. Graphs in $\mathfrak{I}_S$ are in $\mathfrak{I}$, hence have contractible clique complex and are $\mathbb{Z}_2$-acyclic; the converse holds below n by minimality of n. □

2.3. Discs with identified boundary

All the examples found in this paper are K_4-free, so their clique complexes are pure 2-dimensional (every edge lies in a triangle, as will follow from Section 3). For such a complex X let E_3 be the set of edges lying in exactly three triangles (there are no edges in one or in four or more triangles in our examples). Cutting X along E_3 and gluing back the corners around each vertex through the edges that lie in exactly two triangles yields a surface with boundary $\widetilde{X}$ together with a simplicial map $\widetilde{X} \to X$ that is injective on the interior. When $\widetilde{X}$ is a disc, the boundary of $\widetilde{X}$ is mapped onto the cycle (or cycle with pendant edges) formed by E_3, and reading the vertex labels along the boundary gives a *boundary word*. If E_3 is a cycle C of length k and the boundary word runs along C with orientation pattern $a^{\varepsilon_1}a^{\varepsilon_2}a^{\varepsilon_3}$, $\varepsilon_i \in \{\pm 1\}$, each block having length k, then X is homotopy equivalent to the presentation complex of $\langle a \mid a^{\varepsilon_1+\varepsilon_2+\varepsilon_3}\rangle$: contractible for the dunce-hat pattern $a\,a\,a^{-1}$ and a Moore space $M(\mathbb{Z}/3,1)$ for the pattern a^3. This is exactly what happens for the 15-vertex dunce hat D_{15} of [12] and for 48 of the 51 stuck graphs on 14 vertices (Section 6).

3. Reduction to stuck graphs

The number of connected graphs on 13 vertices (50,335,907,869,219 $\approx 5.0 \times 10^{13}$) and on 14 vertices ($\approx 2.9 \times 10^{16}$; OEIS A001349) makes a naive search for the smallest member of $\mathfrak{I}\setminus\mathfrak{I}_S$ impractical: deciding $\mathfrak{I}_S$ exactly costs microseconds to milliseconds per graph. The following observation turns the

search for a smallest counterexample into a search for a much rarer kind of graph.

Definition 3.1. A graph G is *stuck* if it is $\mathbb{Z}_2$-acyclic and has no $\mathfrak{I}_S$-deletable vertex and no $\mathfrak{I}_S$-deletable edge, i.e. $N(v) \notin \mathfrak{I}_S$ for every vertex v and $N(u,v) \notin \mathfrak{I}_S$ for every edge uv (recall that $\emptyset \notin \mathfrak{I}_S$). A graph A with a distinguished vertex u is *almost stuck at* u if it is $\mathbb{Z}_2$-acyclic, has no $\mathfrak{I}_S$-deletable edge, and no vertex other than u is $\mathfrak{I}_S$-deletable (u itself may or may not be).

A stuck graph is trivially outside $\mathfrak{I}_S$: it admits no strong contractible move at all. Conversely:

Lemma 3.2 (Minimality implies stuck)**.** *(a) Let n_0 be the smallest order of a $\mathbb{Z}_2$-acyclic graph not in $\mathfrak{I}_S$. Every $\mathbb{Z}_2$-acyclic graph on n_0 vertices not in $\mathfrak{I}_S$ is obtained from a stuck graph on n_0 vertices by successively adding edges uv with $N(u,v) \in \mathfrak{I}_S$ (transformations (I4)); we say it lies in the* I_4-closure *of that stuck graph. Those with the minimum number of edges are themselves stuck.*

(b) Let n_1 be the smallest order of a graph in $\mathfrak{I} \setminus \mathfrak{I}_S$. Every graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on n_1 vertices lies in the I_4-closure of a stuck graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on n_1 vertices, which is K_4-free if the original graph is. Those with the minimum number of edges (in the whole class, or in the K_4-free class) are stuck.

Lemma 3.3 (Stuck graphs are 2-connected)**.** *Let G be a stuck graph with a cut vertex u, so that $G = A \cup_u B$ with A, B connected, $V(A) \cap V(B) = \{u\}$ and no edges between $A - u$ and $B - u$. Then A and B are almost stuck at u and have at least three vertices. Consequently, if there is no almost stuck graph with at most m vertices, every stuck graph with at most $2m$ vertices is 2-connected.*

Lemma 3.4 (Degree bounds)**.** *Let G be a 2-connected stuck graph on $n \geq 2$ vertices. Then every edge of G lies in a triangle, every common neighbourhood $N(u,v)$ is non-empty, $\delta(G) \geq 4$ and $\Delta(G) \leq n - 2$.*

The proofs are elementary and are given in Appendix A; Fig. 1 illustrates Lemma 3.3.

To test stuckness during an enumeration one would like to replace the condition "$\notin \mathfrak{I}_S$" on neighbourhoods by the cheaper "not $\mathbb{Z}_2$-acyclic". This is legitimate in the range of this paper:

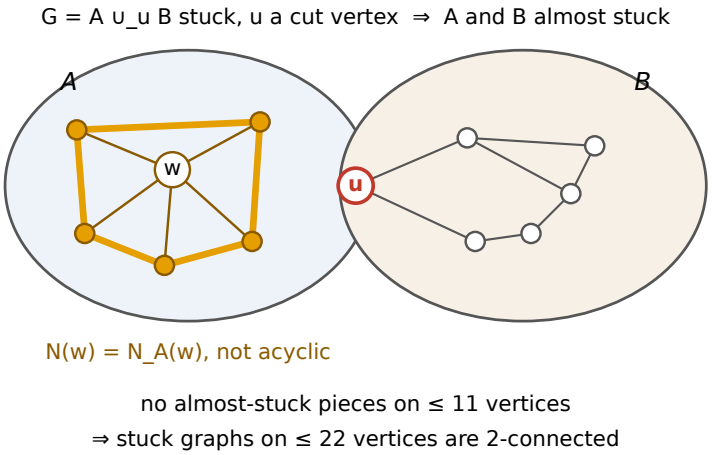


Figure 1: Lemma 3.3: if a stuck graph $G = A \cup_u B$ has a cut vertex u, the neighbourhoods of all vertices and edges of A other than u are computed inside A, so A is almost stuck at u. Since no almost stuck graph has at most 11 vertices (Proposition 3.6), stuck graphs with at most 22 vertices are 2-connected.

Lemma 3.5. *Let G be a $\mathbb{Z}_2$-acyclic graph on $n \leq 14$ vertices without a universal vertex. Then G is stuck if and only if no open neighbourhood $N(v)$ and no non-empty common neighbourhood $N(u, v)$ is $\mathbb{Z}_2$-acyclic. The same holds for "almost stuck" when $n \leq 11$.*

Proof. All neighbourhoods involved have at most $n - 2 \leq 12$ vertices, and for graphs with at most 12 vertices $\mathbb{Z}_2$-acyclic is equivalent to $\mathfrak{I}_S$ (Table 2, Theorem 6.1). □

Proposition 3.6. *There is no almost stuck graph with at most 11 vertices. Consequently every stuck graph with at most 22 vertices is 2-connected.*

Proof. An almost stuck graph in which u is not $\mathfrak{I}_S$-deletable is stuck, hence $\mathbb{Z}_2$-acyclic and outside $\mathfrak{I}_S$; by Table 2 there is no such graph with at most 12 vertices. Algorithm 1 in mode *almost*, which by Lemma 3.5 detects exactly the $\mathbb{Z}_2$-acyclic graphs with no $\mathfrak{I}_S$-deletable edge and exactly one $\mathfrak{I}_S$-deletable vertex, was run on all connected graphs with $3 \leq n \leq 11$ vertices (1,006,700,565 graphs for $n = 11$) and found none. The second statement is Lemma 3.3. □

Theorem 3.7 (Reduction). *(i) If $n_0 \le 22$ is the smallest order of a $\mathbb{Z}_2$-acyclic graph outside $\mathfrak{I}_S$, every such graph on n_0 vertices lies in the I_4-closure of a stuck graph on n_0 vertices.*

(ii) If $n_1 \le 22$ is the smallest order of a graph in $\mathfrak{I} \setminus \mathfrak{I}_S$, every graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on n_1 vertices lies in the I_4-closure of a stuck graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on n_1 vertices, K_4-free if the graph is.

In both cases every stuck graph on n_i vertices is 2-connected with $4 \le \delta \le \Delta \le n_i - 2$, and for $n_i \le 14$ stuckness is decided by $\mathbb{Z}_2$-homology of the neighbourhoods (Lemma 3.5). Hence, to find all graphs of either kind on n_i vertices it suffices to enumerate the 2-connected graphs with these degree bounds, keep the stuck ones, and compute their I_4-closures.

For $n = 13$ the enumeration of 2-connected graphs with $4 \le \delta$ and $\Delta \le 11$ comprises 21,011,415,992,470 graphs, about 2.4 times fewer than all connected graphs; the real gain lies in the stuck filter: the cascade of Section 4 rejects all but a fraction 7.5×10^{-4} of the graphs by looking at cones, the Euler characteristic and a few neighbourhoods, computes the homology of G itself for only 15,693,920,490 graphs, and never runs the exact $\mathfrak{I}_S$ decision during the enumeration. For $n = 14$ we restricted the search to K_4-free graphs (clique complexes of dimension 2); this class contains the dunce hat D_{15} and comprises 64,856,470,381,820 graphs under the same degree constraints. By Theorem 3.7(ii) this determines all K_4-free graphs in $\mathfrak{I} \setminus \mathfrak{I}_S$ on 14 vertices.

4. Algorithms and the computation on YUCA

All algorithms are implemented in a single C program (`src/atascadas.c` in the repository; *atascada* is Spanish for *stuck*) that reads graphs in `graph6` format from the output of `geng` [19]. Graphs have at most 15 vertices, so vertex sets are bitmasks and neighbourhoods are arrays of bitmasks; cliques are enumerated by recursive extension. Independent re-implementations in pure Python of the exact $\mathfrak{I}_S$ decision and of integer homology (`tools/`) are used for the certificates (Section 8).

4.1. Homology and the stuck filter

$\mathbb{Z}_2$-homology of $\Delta(H)$ for a small graph H is computed by Gaussian elimination over GF(2) of the boundary matrices, with simplices up to dimension $\omega(H) - 1$. The Euler characteristic $\chi(H) = \sum_k (-1)^{k-1} c_k(H)$ is

Algorithm 1: STUCK(G) — filter cascade

Input: a graph G on n vertices (bitmask adjacency)
Output: **true** iff G is stuck

foreach *vertex* v **do**
  if $|N(v)| < 2$ ***or*** $N(v)$ *is a cone* **then return false**;
if $\chi(G) \neq 1$ **then return false**;
foreach *vertex* v **do**
  if $N(v)$ *is* $\mathbb{Z}_2$*-acyclic* **then return false**;
foreach *edge* uv **do**
  if $N(u, v) \neq \emptyset$ ***and*** $N(u, v)$ *is* $\mathbb{Z}_2$*-acyclic* **then return false**;
if G *is not* $\mathbb{Z}_2$*-acyclic* **then return false**;
return true

obtained from the clique enumeration and is a much cheaper necessary condition ($\chi = 1$) for $\mathbb{Z}_2$-acyclicity. Integer homology of the (few) candidate graphs is computed by exact Smith normal form of the boundary matrices over $\mathbb{Z}$; for the 14-vertex graphs the matrices have at most 45×32 entries and the computation is instantaneous.

Algorithm 1 decides whether G is stuck by a cascade of tests of increasing cost; the tests are ordered so that the overwhelming majority of graphs is rejected early.

The acyclicity test for a neighbourhood H is itself a cascade: H disconnected $\Rightarrow$ not acyclic; H 0-dismantlable (greedy cone-vertex removal) $\Rightarrow$ acyclic; $\chi(H) \neq 1 \Rightarrow$ not acyclic; otherwise compute $\widetilde{H}_*(H; \mathbb{Z}_2)$. The *almost stuck* variant used for Proposition 3.6 replaces the vertex loop by a count of vertices with acyclic neighbourhood, accepting exactly one.

4.2. Exact decision of $\mathfrak{I}_S$, $\mathfrak{I}_{VS}$ and k-Dis

Since a greedy sequence of deletions may get stuck in a graph of $\mathfrak{I}_S$ ([12, Fig. 3]), we decide $\mathfrak{I}_S$ by exhaustive search over deletion sequences (Algorithm 2). The search is memoised on the pair (vertex set, edge set) of the current graph; a 0-dismantlable graph is accepted immediately ([12, Lemma 19]); the neighbourhood conditions are decided recursively by the same procedure. For graphs with at most 15 vertices the memoised search terminates in milliseconds for the graphs appearing here. $\mathfrak{I}_{VS}$ and k-Dis are

Algorithm 2: InIS(G) — exact membership in $\mathfrak{I}_S$

Input: a graph $G = (V, E)$
Output: **true** iff $G \in \mathfrak{I}_S$ (and, on success, a deletion sequence)

```
if |V| = 0 then return false;
if |V| = 1 or G is 0-dismantlable then return true;
if (V, E) is memoised then return the stored answer;
foreach vertex v with N(v) ≠ ∅ and InIS(N(v)) do
    if InIS(G − v) then memoise and return true;
foreach edge uv with N(u, v) ≠ ∅ and InIS(N(u, v)) do
    if InIS(G − uv) then memoise and return true;
memoise and return false
```

Algorithm 3: ICert(G, d) — certificate of I-contractibility by $\leq d$ additions

```
Q ← {(G, ∅)};
for ℓ = 0, ..., d do
    foreach (H, L) ∈ Q do
        if InIS(H) then return (L, deletion sequence of H);
    Q ← {(H + uv, L · uv) : (H, L) ∈ Q, uv ∉ E(H), N_H(u, v) ∈ I_S},
        one representative per isomorphism class;
return undecided
```

decided by dynamic programming over the 2^n induced subgraphs: a set S is reducible if $|S| = 1$ or some $v \in S$ has $N_{G[S]}(v)$ in the relevant class and $S \setminus \{v\}$ reducible.

4.3. Certificates of I-contractibility and I_4*-closures*

To certify that a stuck graph G is I-contractible we search breadth-first for a short sequence of edge additions uv with $N(u, v) \in \mathfrak{I}_S$ (transformations (I4), legitimate since $\mathfrak{I}_S \subseteq \mathfrak{I}$) leading to a graph in $\mathfrak{I}_S$ (Algorithm 3); the certificate consists of the added edges together with the deletion sequence of the final graph produced by Algorithm 2, and can be checked by hand or by the Python tools. For all three examples on 14 vertices a single edge suffices.

The I_4*-closure* of a stuck graph (Lemma 3.2(b)) is computed by the same breadth-first expansion without the stopping rule, recording for every su-

pergraph whether it is in $\mathfrak{I}_S$; isomorphism reduction uses `nauty`'s canonical labelling. For the 51 stuck graphs on 14 vertices this produces all $\mathbb{Z}_2$-acyclic graphs on 14 vertices outside $\mathfrak{I}_S$ that lie over a K_4-free stuck graph (Section 6.2).

4.4. Unfolding into a disc

For a K_4-free candidate we cut $\Delta(G)$ along the edges lying in three triangles and glue the corners back through the edges in two triangles (Section 2); when the result is a disc we draw it with a Tutte embedding (boundary on a regular polygon, interior vertices at the barycentres of their neighbours) and read off the boundary word. This produces the figures of Section 6 and classifies the 51 stuck graphs by the pattern of their boundary identification.

4.5. The distributed search

The enumeration was run on the YUCA cluster of ACARUS (Universidad de Sonora): CPU partition of 24 nodes with two 32-core AMD EPYC 9354 processors (64 cores, no hyper-threading) and 1 TB of RAM each, Rocky Linux 9.6, SLURM 24.11, Lustre file system, SLURM with a per-project quality-of-service limit of 500 cores, `gcc` 11, `nauty` 2.7r4 (and `nauty` 2.8 compiled locally for the K_4-free option `-k` of `geng`). Algorithm 4 and Fig. 2 describe the workflow; the exact scripts are in the repository (`yuca/`).

For $n = 14$ a second wrapper, submitted under a second project account, processed the slices in descending order sharing the same registries, so that the two fronts met in the middle; 692 slices were processed by both fronts with identical output, and coverage of all $M = 2{,}000{,}000$ slices was verified after completion. Table 1 summarises the runs. The per-slice cost is dominated by `geng` (about $2.7\,\mu$s per graph); Algorithm 1 costs about $0.8\,\mu$s per graph on average because most graphs fail the cone test.

4.6. Validation

The program reproduces every entry of Table 1 of [12] for $n \le 11$ (Section 5); it classifies the dunce hat D_{15} of [12, Fig. 2A] as $\mathbb{Z}_2$-acyclic, stuck and not in $\mathfrak{I}_S$, and the graph of [12, Fig. 2B] as a member of $\mathfrak{I}_S$; and its exact $\mathfrak{I}_S$ decision agrees with the independent Python implementation on all graphs reported here. The row $n = 12$ agrees with an independent computation by the authors' group (unpublished, 2025) in every cell that both computations cover. Integer homology groups were computed with our own Smith normal form routine and cross-checked with `sympy` and with Wolfram Mathematica for the key graphs.

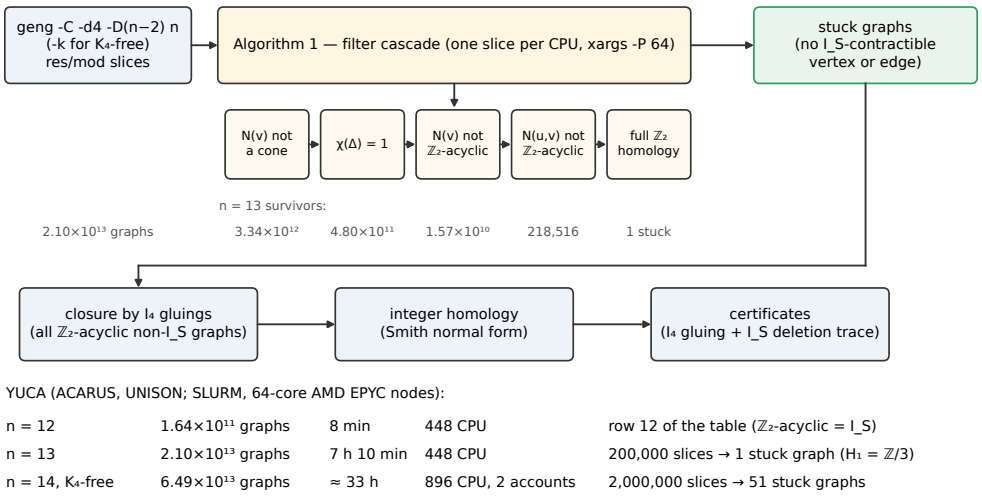


Figure 2: The computational pipeline: `geng` enumeration in `res/mod` slices, the filter cascade of Algorithm 1 (with the survivor counts of the $n = 13$ run), I_4-closures, integer homology and certificates; bottom: the three cluster runs.

5. The extended cardinality table

Table 2 lists, for connected graphs on $n \leq 12$ vertices, the number of graphs in each of the classes of Section 2. The rows $n \leq 11$ coincide cell by cell with Table 1 of [12]; the column $\mathfrak{I}$ is new (it was obtained by certifying every $\mathbb{Z}_2$-acyclic graph outside $\mathfrak{I}_S$ with Algorithm 3, which for $n \leq 12$ is vacuous since every $\mathbb{Z}_2$-acyclic graph is in $\mathfrak{I}_S$). The row $n = 12$ was computed on the cluster in 8 minutes (Table 1) and gives the new value $|\mathfrak{I}_{VS}(12)| = 22{,}666{,}525{,}713$; the columns 3-Dis, 2-Dis, 1-Dis and (-1)-Dis for $n = 12$ were not part of that run and are being computed together with the row $n = 13$.

Several patterns in the table are worth recording.

Observation 5.1. (i) For $n \leq 12$ every $\mathbb{Z}_2$-acyclic graph is strong I-contractible: $\mathbb{Z}_2$-acyclic $= \mathfrak{I} = \mathfrak{I}_S$. The first exceptions occur at $n = 13$ and are the three torsion graphs of Theorem 6.1; the first members of $\mathfrak{I} \setminus \mathfrak{I}_S$ occur at $n = 14$ (Theorem 6.4).

(ii) $\mathfrak{I}_{VS} =$ 2-Dis $=$ 3-Dis for all $n \leq 11$, and $|\mathfrak{I}_{VS}(12)|$ agrees with the value of $|2\text{-Dis}(12)|$ obtained in an independent (unpublished, 2025)

Algorithm 4: Distributed search on the cluster

Input: n; number of slices M; number of array tasks T; `geng` options Ω; mode (stuck / table)

A one-core wrapper job compiles the program, creates the registries `listas_t`, `enc_t` ($t = 0, \dots, T-1$) and submits a SLURM array of T tasks with 64 cores each;

foreach *task* t *in parallel* **do**

  foreach *slice* $r \equiv t \pmod T$, $0 \le r < M$, *with* 64 *slices in flight (`xargs -P 64`)* **do**

    if $r \in$ *`listas_t`* **then** skip (resumable);

    run `geng` Ω n `-q` r/M `|` `atascadas --modo` mode, writing the output to a per-slice file;

    under a file lock, append the output to `enc_t` and r to `listas_t`;

The wrapper polls progress every 60 s; when all tasks end it sums the counters, verifies that $\bigcup_t$ `listas_t` $= \{0, \dots, M-1\}$, lists duplicated slices, and records `sacct` accounting;

computation by the authors; thus $\mathfrak{I}_{VS} = 2$-Dis up to 12 vertices, and a graph in $\mathfrak{I}_{VS}$ that is not 2-dismantlable, if any, has at least 13 vertices.

(iii) $\mathfrak{I}_S \neq \mathfrak{I}_{VS}$ first at $n = 11$ (twelve graphs; the smallest members of $\mathfrak{I}_S \setminus \mathfrak{I}_{VS}$ of [12]), and $|\mathfrak{I}_S(12)| - |\mathfrak{I}_{VS}(12)| = 11{,}718$.

(iv) $\chi = 1$ but not $\mathbb{Z}_2$-acyclic first at $n = 7$ (one graph), and the gap grows to 9,816,037,137 graphs at $n = 12$.

6. The minimal examples

Throughout, vertices are labelled $0, \dots, n-1$ as in the `graph6` strings given in the text and in the repository, so that every statement can be checked mechanically.

6.1. Thirteen vertices: odd torsion

Theorem 6.1. *Every $\mathbb{Z}_2$-acyclic graph on at most 12 vertices is strong I-contractible. On 13 vertices there are exactly three $\mathbb{Z}_2$-acyclic graphs that are not strong I-contractible, namely*

Table 1: Searches performed. "graphs" is the number of graphs produced by `geng` and examined; the filter counters are the numbers of graphs passing the cone test, the Euler characteristic test and the neighbourhood tests of Algorithm 1.

n	class (`geng` options)	mode	graphs	pass cone	pass χ	pass N	stuck
12	connected (`-c`)	table	164,059,830,476	–	–	–	0
13	`-C -d4 -D11`	stuck	21,011,415,992,470	3,335,926,431,910	480,204,455,162	15,693,920,490	1
14	`-k -C -d4 -D12`	stuck	64,856,470,381,820	48,303,369,131,820	1,462,235,073,536	108,022,349,478	51

n	slices M	cores	wall time	SLURM arrays
12	4,000	448	8 min	2079049, 2079051
13	200,000	448	7 h 10 min	2079058, 2079059
14	2,000,000	896	$\approx$ 33 h	2079102 (p029), 2079156 (p011)

$$G_{13} = \texttt{L?qa``\{wttym\~{}o} \text{ (40 edges)}, \qquad G_{13} + \{4,7\} \text{ (41)}, \qquad G_{13} + \{4,7\} + \{5,6\} \text{ (42)}.$$

G_{13} is the unique stuck graph on 13 vertices and the other two form its I_4-closure. For all three, $H_*(\,\cdot\,;\mathbb{Z})$ is $\mathbb{Z}/3$ in degree 1 and zero otherwise; in particular none of them is I-contractible, and G_{13} is a flag triangulation of the Moore space $M(\mathbb{Z}/3,1)$ with 28 triangles.

Proof. The first statement is the equality of the columns "$\mathbb{Z}_2$-acyclic" and "$\mathfrak{I}_S$" in Table 2 for $n \le 12$ (exact decision of $\mathfrak{I}_S$ for each $\mathbb{Z}_2$-acyclic graph). For $n = 13$, Theorem 3.7 applies: the enumeration of all 2-connected graphs with $4 \le \delta$, $\Delta \le 11$ (Table 1) found exactly one stuck graph, G_{13}, and its I_4-closure (Algorithm 3 without stopping rule) consists of the two graphs listed, both outside $\mathfrak{I}_S$ (every edge uv with $N(u,v) \in \mathfrak{I}_S$ was added; the three graphs are pairwise non-isomorphic since they have different numbers of edges). The homology is computed by Smith normal form (file `certificates/integer_homology.txt`) and was cross-checked with `sympy` and Mathematica. Since graphs in $\mathfrak{I}$ have contractible, hence $\mathbb{Z}$-acyclic, clique complexes, none of the three is in $\mathfrak{I}$. □

The structure of G_{13} is shown in Fig. 3. Its clique complex is pure 2-dimensional; exactly four edges lie in three triangles and they form the induced 4-cycle $C = 8$–10–9–11. Cutting along C yields a disc whose boundary of length 12 is mapped onto C with the word $(10\,9\,11\,8)^3$, that is, with the pattern a^3: $\Delta(G_{13}) \simeq M(\mathbb{Z}/3,1)$. The automorphism group has order 8, generated by $(0\,1\,3\,2)(4\,6\,7\,5)(8\,10\,9\,11)$ and $(0\,1)(2\,3)(5\,6)(8\,10)(9\,11)$; the degree sequence is $5^4\,6^4\,7^4\,8$. The open neighbourhoods of the nine vertices off

Table 2: Numbers of connected graphs on n vertices in each class. Rows $n \le 11$ agree with [12, Table 1]. †For $n = 14$ only the K_4-free class has been searched.

n	connected	$\chi = 1$	$\mathbb{Z}_2$-acyclic	$\mathfrak{I}$	$\mathfrak{I}_S$	$\mathfrak{I}_{VS}$	3-Dis	2-Dis	1-Dis	0-Dis	(-1)-Dis
3	2	2	2	2	2	2	2	2	2	2	2
4	6	5	5	5	5	5	5	5	5	5	4
5	21	16	16	16	16	16	16	16	16	16	11
6	112	68	68	68	68	68	68	68	68	68	34
7	853	404	403	403	403	403	403	403	403	403	156
8	11,117	3,837	3,798	3,798	3,798	3,798	3,798	3,798	3,798	3,791	1,044
9	261,080	68,458	66,167	66,167	66,167	66,167	66,167	66,167	66,167	65,561	12,346
10	11,716,571	2,571,263	2,343,148	2,343,148	2,343,148	2,343,148	2,343,148	2,343,148	2,343,148	2,258,314	274,668
11	1,006,700,565	204,131,202	166,978,401	166,978,401	166,978,401	166,978,389	166,978,389	166,978,389	166,978,389	151,084,672	12,005,168
12	164,059,830,476	32,482,574,568	22,666,537,431	22,666,537,431	22,666,537,431	22,666,525,713	–	–	–	18,510,237,804	–
13	50,335,907,869,219	*computation in progress*; $\mathbb{Z}_2$-acyclic $\setminus \mathfrak{I}_S = 3$, $\mathfrak{I} \setminus \mathfrak{I}_S = 0$ (Theorem 6.1)									
14†		*computation in progress*; stuck $\mathbb{Z}_2$-acyclic $\setminus \mathfrak{I}_S = 51$, $\mathfrak{I} \setminus \mathfrak{I}_S \ge 29$ (Theorem 6.4)									

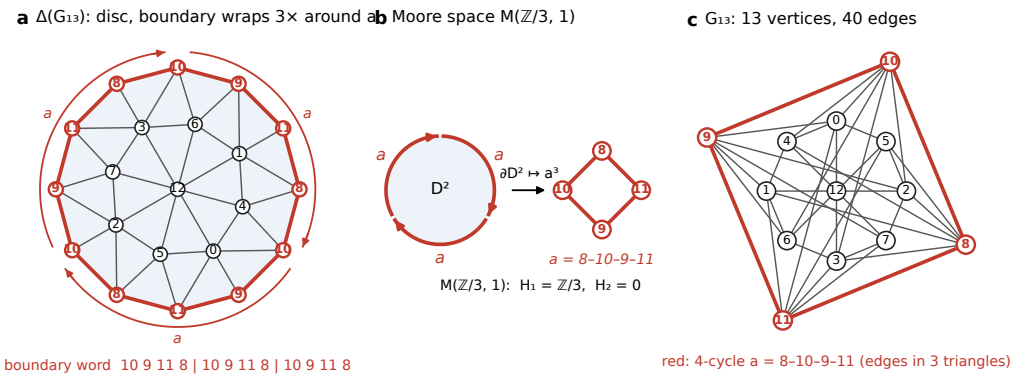


Figure 3: The graph G_{13} on 13 vertices. (a) Its clique complex cut open along the four edges lying in three triangles (8–10, 10–9, 9–11, 11–8) is a disc with 28 triangles and 21 vertices whose boundary word is $(10\,9\,11\,8)^3$: the boundary winds three times around the 4-cycle in the same direction. (b) The resulting space is the Moore space $M(\mathbb{Z}/3, 1)$. (c) A plane drawing of G_{13} with vertex 12 (degree 8) at the centre, its neighbourhood C_8 around it and the 4-cycle in red.

C are induced cycles (C_6 for $0, 1, 2, 3$; C_5 for $4, 5, 6, 7$; C_8 for 12) and those of the four vertices of C have seven vertices and eight edges (a cycle with a chord path), so no vertex is $\mathfrak{I}_S$-deletable; no common neighbourhood $N(u, v)$ is $\mathbb{Z}_2$-acyclic either (each is non-empty, since every edge lies in a triangle, and has at least two vertices and no cone point), so no edge is deletable—this is what "stuck" means.

Remark 6.2 (Torsion in small flag complexes). Adamaszek [16] classified the flag complexes with at most 12 vertices whose first homology has torsion: there are none with at most 10 vertices, 4 on 11 vertices and 394 on 12, and all of them have $H_1 \cong \mathbb{Z}/2$ (they contain $\mathbb{RP}^2$-like configurations). Theorem 6.1 shows that 13 vertices suffice for $\mathbb{Z}/3$-torsion in a flag complex, and the three graphs above are therefore the smallest flag complexes with odd torsion. (Non-flag triangulations of $M(\mathbb{Z}/3, 1)$ exist with far fewer vertices; the flag condition is the obstruction.) We do not know whether there are other 13-vertex flag complexes with odd torsion that are not $\mathbb{Z}_2$-acyclic, since our search only covered the $\mathbb{Z}_2$-acyclic ones.

Proposition 6.3. *There is no flag triangulation of the dunce hat on* 13 *vertices whose* 1*-skeleton is* $\mathbb{Z}_2$*-acyclic and not strong I-contractible; in fact there is no graph on at most* 13 *vertices in* $\mathfrak{I} \setminus \mathfrak{I}_S$.

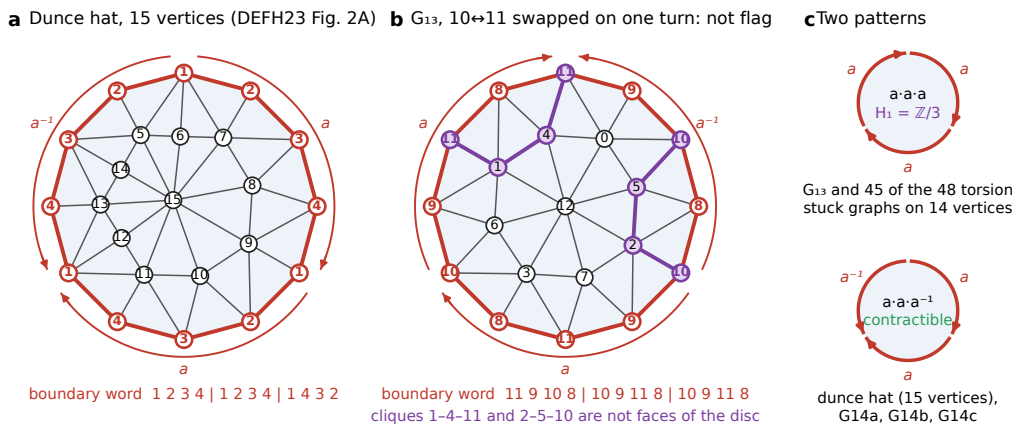


Figure 4: (a) The 15-vertex flag dunce hat D_{15} of [12, Fig. 2A] unfolded: a disc with 23 vertices whose boundary of length 12 runs along the 4-cycle 1–2–3–4 with the word 1 2 3 4 | 1 2 3 4 | 1 4 3 2, i.e. the pattern $a\,a\,a^{-1}$ (labels $1,\dots,15$ of [12]; the `graph6` label i corresponds to the label $(1,4,13,12,11,3,14,15,10,5,2,6,7,8,9)_i$, so this cycle is 0–10–5–1 in `graph6` labels). (b) Reversing one of the three passes of the boundary of G_{13} (exchanging $10 \leftrightarrow 11$ in one block) gives a dunce hat, but its 1-skeleton contains the cliques 1–4–11 and 2–5–10 (purple) that are not faces of the disc, so the complex is not flag; the resulting graph is vertex-strong I-contractible. (c) The two identification patterns occurring among the stuck graphs of this paper.

Proof. By Theorem 6.1 the only $\mathbb{Z}_2$-acyclic graphs on ≤ 13 vertices outside $\mathfrak{I}_S$ are the three torsion graphs, none of which is contractible. □

It is instructive to see what goes wrong when one tries to turn G_{13} into a dunce hat (Fig. 4(b)): reversing the orientation of one block of the boundary word changes the pattern to $a\,a\,a^{-1}$ and the resulting complex is contractible, but two triples of pairwise adjacent vertices are no longer faces, so the complex is not the clique complex of its 1-skeleton, and that 1-skeleton (the graph `L?qa``{wtlzM~o`, canonical form `LtDzQNKmpKXT`V`) contains the two copies $\{1,4,8,11\}$, $\{2,5,8,10\}$ of K_4 and is in $\mathfrak{I}_{VS}$.

6.2. Fourteen vertices: the first examples of $\mathfrak{I} \setminus \mathfrak{I}_S$

Theorem 6.4. *There is no graph on at most* 13 *vertices that is I-contractible but not strong I-contractible. Among K_4-free graphs on* 14 *vertices there are exactly* 51 *stuck graphs. Exactly three of them have trivial integer homology,*

$$G^a_{14} = \texttt{M?b@bFHNEk@~/k}\}\texttt{U?}, \qquad G^b_{14} = \texttt{M?B@thku?~Lylk}\}\texttt{[?},$$
$$G^c_{14} = \texttt{M?r@d`ifbiFfo}\}\texttt{~o?},$$

with 44, 45 *and* 45 *edges and* 31, 32 *and* 32 *triangles; each is I-contractible but not strong I-contractible. The remaining* 48 *have* $H_1(\,\cdot\,;\mathbb{Z}) \cong \mathbb{Z}/3$. *Consequently the smallest graphs in* $\mathfrak{I} \setminus \mathfrak{I}_S$ *have exactly* 14 *vertices, and* G^a_{14} *has the minimum number of edges among the* K_4*-free ones.*

Proof. The first statement is Proposition 6.3. The enumeration of K_4-free 2-connected graphs on 14 vertices with $4 \le \delta$, $\Delta \le 12$ (Table 1; 2,000,000 slices, all accounted for) returned 51 stuck graphs, pairwise non-isomorphic by canonical labelling; their integer homology was computed by Smith normal form. For G^a_{14}, the common neighbourhood of the non-adjacent vertices 0 and 1 is the induced path 8–12–5–10–13, which is in $\mathfrak{I}_S$, so $G^a_{14}+\{0,1\}$ is obtained from G^a_{14} by a transformation (I4); Algorithm 2 finds a sequence of 18 strong deletions reducing $G^a_{14} + \{0,1\}$ to K_1 (Appendix C), so $G^a_{14} + \{0,1\} \in \mathfrak{I}_S \subseteq \mathfrak{I}$ and $G^a_{14} \in \mathfrak{I}$. Similarly $N_{G^b_{14}}(0,3) = 7$–13–9–12 and $N_{G^c_{14}}(2,3) =$ 6–13–7–10–11 are induced paths and $G^b_{14}+\{0,3\}$, $G^c_{14}+\{2,3\}$ are in $\mathfrak{I}_S$ with 19 and 18 deletions. The three graphs are stuck, so not in $\mathfrak{I}_S$. Finally, a K_4-free graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on 14 vertices with fewer than 44 edges would, by Lemma 3.2(b) (edge deletions preserve K_4-freeness), lie in the I_4-closure of a stuck K_4-free graph with at most 43 edges; all such stuck graphs have $H_1 \cong \mathbb{Z}/3$, and (I4) additions preserve the homotopy type, a contradiction. □

Figures 5–7 show the three graphs. All three clique complexes are pure 2-dimensional flag triangulations of a disc with its boundary identified along a short induced cycle with the dunce-hat pattern $a\,a\,a^{-1}$ (Table 3); they are therefore homotopy equivalent to the dunce hat and contractible but, like the dunce hat, presumably not collapsible (see Section 7.3). Compared with D_{15} they use a longer identification cycle (5, 4 and 6 instead of 4), fewer edges (44–45 instead of 46) and a more homogeneous degree sequence (degrees between 5 and 8 instead of between 4 and 10). In all four graphs the vertices outside the identification cycle have an induced cycle as open neighbourhood (so their links are circles: these are interior points of the surface), and the vertices on the cycle have a neighbourhood with one more independent cycle.

The 48 torsion examples are listed in Appendix B (Table B.4) and shown in Fig. 9. They have 43 (25 graphs), 44 (18), 45 (4) or 47 (1) edges. For 45 of them the unfolding is a disc whose boundary is identified with a cycle of length 4 (25 graphs), 5 (18) or 6 (2) with the pattern a^3, as for G_{13}; two unfold to an annulus and one (the graph with 47 edges, whose edges in three triangles form two disjoint 4-cycles) to a disconnected surface, so their $\mathbb{Z}/3$ does not come from a single disc identification.

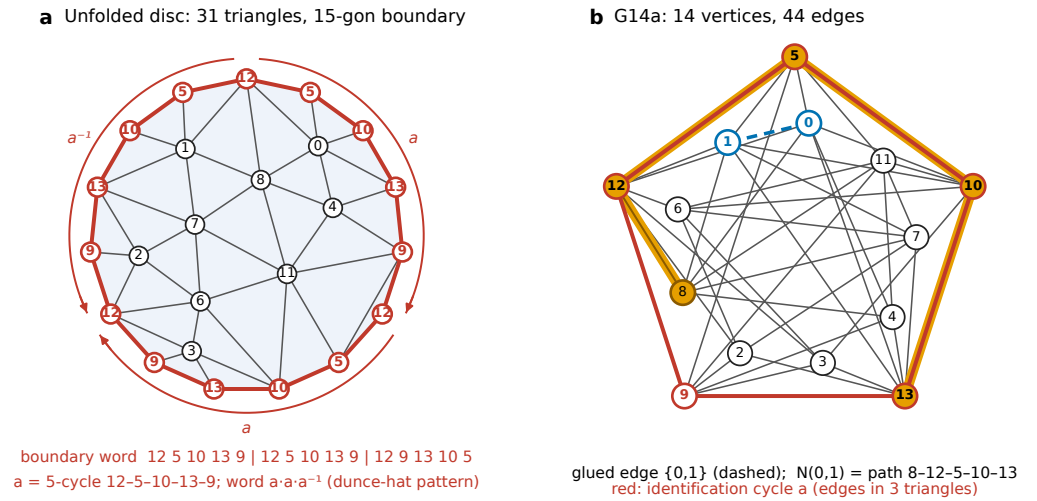


Figure 5: G_{14}^{a} (44 edges, 31 triangles, trivial automorphism group). (a) Unfolded disc with 24 vertices; the boundary of length 15 is identified with the 5-cycle 5–10–13–9–12 with the dunce-hat pattern $a\,a\,a^{-1}$ (boundary word 12 5 10 13 9 | 12 5 10 13 9 | 12 9 13 10 5). (b) Plane drawing; the dashed edge $\{0,1\}$ is the (I4) addition certifying I-contractibility, and its common neighbourhood, the induced path 8–12–5–10–13, is highlighted.

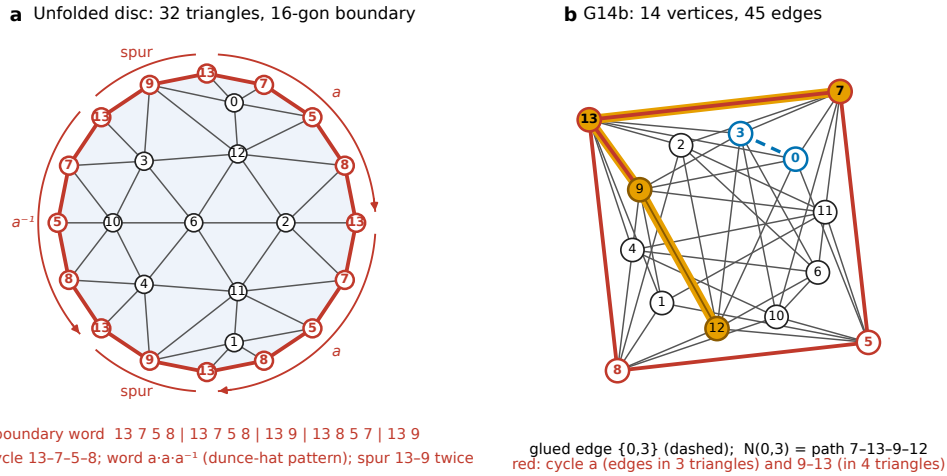


Figure 6: G_{14}^{b} (45 edges, 32 triangles, $|\mathrm{Aut}| = 2$). The boundary of the unfolded disc (25 vertices, length 16) is identified with the 4-cycle 5–7–13–8 with pattern $a\,a\,a^{-1}$, and additionally runs back and forth along the pendant edge 13–9, which also lies in three triangles. The (I4) edge is $\{0,3\}$ with $N(0,3) = 7$–13–9–12.

Table 3: The three minimal examples and the dunce hat of [12]. E_3 = edges in three triangles; disc = number of vertices of the unfolded disc; k = length of the identification cycle.

	n	m	triangles	degrees	$\lvert E_3\rvert$	$\lvert\mathrm{Aut}\rvert$	disc	k	pattern
D_{15}	15	46	32	$4^3\,5^2\,6^7\,9^2\,10$	4	2	23	4	$a\,a\,a^{-1}$
G_{14}^a	14	44	31	$5^3\,6^6\,7^3\,8^2$	5	1	24	5	$a\,a\,a^{-1}$
G_{14}^b	14	45	32	$5^2\,6^5\,7^6\,8$	4 (+1 pendant)	2	25	4	$a\,a\,a^{-1}$
G_{14}^c	14	45	32	$6^9\,7^4\,8$	6	2	26	6	$a\,a\,a^{-1}$

The I_4*-closures.* By Theorem 3.7(ii), every K_4-free graph in $\mathfrak{I}\setminus\mathfrak{I}_S$ on 14 vertices lies in the I_4-closure of one of the 51 stuck graphs (the closure itself may leave the K_4-free class). Computing all 51 closures gives 2,271 pairwise non-isomorphic $\mathbb{Z}_2$-acyclic graphs on 14 vertices outside $\mathfrak{I}_S$, of which 29 are contractible, hence in $\mathfrak{I}\setminus\mathfrak{I}_S$, and 2,242 have $H_1\cong\mathbb{Z}/3$ (file `data/acyclic_not_strong_14_closure.tsv` in the repository); 8 of the 29 are K_4-free, with 44–47 edges. Hence the K_4-free graphs on 14 vertices in $\mathfrak{I}\setminus\mathfrak{I}_S$ are exactly these 8 graphs. (The closures do not exhaust the $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$ on 14 vertices: adding to one of the three 13-vertex torsion graphs a new vertex whose neighbourhood is in $\mathfrak{I}_S$ produces further graphs with $H_1\cong\mathbb{Z}/3$ outside $\mathfrak{I}_S$, which are not stuck and not in these closures. Lemma 3.2(a) does not apply at $n=14$ since $n_0=13$.)

6.3. Relation with the dunce hat of [12]

The graph D_{15} (Fig. 4(a)) is also stuck, with 46 edges, 32 triangles and the identification pattern $a\,a\,a^{-1}$ along the 4-cycle (the 4-cycle 1–2–3–4 in the labels of [12, Fig. 2A]; the label map is recorded in `tools/dunce_hat.py`). It belongs to the same family as $G_{14}^a, G_{14}^b, G_{14}^c$: flag discs with identified boundary. It is not minimal because the same identification can be realised with one vertex fewer by letting the boundary wind around a 5- or 6-cycle, which spreads the three boundary passes over more edges and allows a more uniform (and smaller) vertex set. Adding the edge $\{9,12\}$ to D_{15} (edge $\{5,7\}$ of [12, Fig. 2B]) produces a graph in $\mathfrak{I}_S$ with a 16-step deletion sequence (Appendix C), exactly as adding $\{0,1\}$ to G_{14}^a does.

7. Consequences and open problems

7.1. Conjecture 15 of [12]

In a strong I-contractible graph the order of deletions matters [12, Fig. 3]; Theorem 16 of [12] shows that for vertex-strong I-contractible graphs a bad choice of an $\mathfrak{I}_{VS}$-deletable vertex first occurs at 12 vertices. For edges, Conjecture 15 of [12] proposed that the first bad choice of an $\mathfrak{I}_S$-deletable edge occurs in the 15-vertex graph of their Fig. 2B (the dunce hat D_{15} plus one edge).

Theorem 7.1. *The smallest graph G (in terms of vertices) such that (1) $G \in \mathfrak{I}_S$ and (2) G has an edge $\{v, w\}$ with $N(v, w) \in \mathfrak{I}_S$ and $G - \{v, w\} \notin \mathfrak{I}_S$ has 14 vertices. The graphs $G^a_{14} + \{0, 1\}$, $G^b_{14} + \{0, 3\}$ and $G^c_{14} + \{2, 3\}$ (`M_b@bFHNEk@~/k}U?`, `MCB@thku?~Lylk}[?`, `M@r@d`ifbiFfo}~o?`) are such graphs, and there are 32 pairwise non-isomorphic such graphs in the I_4-closure of G^a_{14} alone.*

Proof. $G^a_{14} + \{0, 1\} \in \mathfrak{I}_S$ with $N(0, 1) \in \mathfrak{I}_S$ and $(G^a_{14} + \{0, 1\}) - \{0, 1\} = G^a_{14} \notin \mathfrak{I}_S$ by Theorem 6.4; likewise for the other two. For minimality, if G on $n \le 13$ vertices satisfied (1) and (2), then $G - \{v, w\}$ would be $\mathbb{Z}_2$-acyclic (the deletion is a homotopy equivalence since $\Delta(N(v, w))$ is contractible) and not in $\mathfrak{I}_S$, so $n = 13$ and $G - \{v, w\}$ is one of the three graphs of Theorem 6.1, by Lemma 3.2(a) a graph in the I_4-closure of G_{13}; but G itself would then be in that closure, which contains no member of $\mathfrak{I}_S$. The count of 32 is by direct computation of the closure of G^a_{14} (`data/`). □

Thus "the order matters" for $\mathfrak{I}_S$-deletable edges first at 14 vertices, two more than for $\mathfrak{I}_{VS}$-deletable vertices.

7.2. The conjectures of [17]

Frías-Armenta [17] gave a 13-vertex counterexample to an axiom used by Ivashchenko and isolated three statements of [2] whose published proofs relied on it, restating them as conjectures: (Conjecture 3) every $G \in \mathfrak{I}$ on n vertices can be transformed into the complete graph K_n by additions of edges uv with $N(u, v) \in \mathfrak{I}$; (Conjecture 4) if $G \in \mathfrak{I}$ and $G - v \in \mathfrak{I}$ then $N(v) \in \mathfrak{I}$; (Conjecture 5) if $G \in \mathfrak{I}$ then $N(v)$ and $G - v$ are I-homotopy equivalent. Our examples are natural test cases, being the smallest graphs where $\mathfrak{I}$ and $\mathfrak{I}_S$ differ, but they do not refute these conjectures.

Proposition 7.2. *Each of G_{14}^a, G_{14}^b, G_{14}^c and D_{15} can be transformed into a cone, hence into K_n, by 11, 12, 12 and 8 additions of edges uv with $N(u,v) \in \mathfrak{I}_S$ respectively (sequences in Appendix C). For every vertex v of these four graphs, $\widetilde{H}_*(G-v;\mathbb{Z}) \cong \widetilde{H}_*(N(v);\mathbb{Z})$, and no $G-v$ is contractible.*

The second statement is not a coincidence: if $\Delta(G)$ is contractible then $\Delta(G) = \Delta(G-v) \cup \mathrm{cone}(\Delta(N(v)))$ with intersection $\Delta(N(v))$, and the Mayer–Vietoris sequence gives $\widetilde{H}_k(G-v) \cong \widetilde{H}_k(N(v))$ for all k. Hence Conjectures 4 and 5 cannot be refuted by homological invariants on graphs in $\mathfrak{I}$; a counterexample would have to be detected by the fundamental group or by simple homotopy type. (By [5], $\mathfrak{I}$-homotopy equivalence of graphs is simple homotopy equivalence of clique complexes, so Conjecture 5 asks whether $\Delta(G-v)$ and $\Delta(N(v))$ are simple homotopy equivalent whenever $\Delta(G)$ is contractible.)

7.3. Open problems

Question 7.3. Is every graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on 14 vertices K_4-free? Equivalently, is the list of eight graphs of Section 6.2 complete?

The full class of 2-connected graphs on 14 vertices with $4 \le \delta \le \Delta \le 12$ has about 10^{16} members, two to three orders of magnitude beyond the present computation. Two routes seem feasible: enumerating only graphs with a bounded number of copies of K_4, or growing the stuck graphs on 13 and 14 vertices by one vertex and testing stuckness of the extensions.

Conjecture 7.4. *Every stuck graph whose clique complex is 2-dimensional unfolds, after cutting along the edges in three triangles, to a disc whose boundary is identified with an induced cycle C of G with the pattern $a\,a\,a^{-1}$ (contractible case) or a^3 (torsion case), possibly with pendant edges traversed back and forth.*

Among the 51 stuck K_4-free graphs on 14 vertices, 48 satisfy the conjecture; the three exceptions (two annuli and one disconnected unfolding) all have 45 or 47 edges and torsion, and it may be that the conjecture holds for stuck graphs of minimum size.

Question 7.5. Are the clique complexes of G_{14}^a, G_{14}^b, G_{14}^c collapsible? Conjecture 27 of [10] asserts that a flag complex is collapsible if and only if its 1-skeleton is in $\mathfrak{I}_S$; since the three graphs are not in $\mathfrak{I}_S$, either they are non-collapsible (as the dunce hat is) or they refute that conjecture.

Question 7.6. Theorem 6.1 and Adamaszek's results [16] give 13 as the smallest number of vertices of a flag complex with odd torsion in H_1 *among* $\mathbb{Z}_2$*-acyclic flag complexes*. Is 13 also the minimum without the $\mathbb{Z}_2$-acyclicity assumption, and which torsion groups occur on 13 and 14 vertices?

Question 7.7. Table 2 shows $\mathfrak{I}_{VS} = 2\text{-Dis} = 3\text{-Dis}$ for $n \leq 12$. Are there graphs in $\mathfrak{I}_{VS}$ that are not 2-dismantlable, and what is the smallest one?

Finally, Question 28 of [12] asked whether every graph with contractible clique complex is I-contractible; this is now the theorem of Dochtermann and Matsushita [5]. Our results show that the analogous question with "$\mathbb{Z}_2$-acyclic" in place of "contractible" has a negative answer already at 13 vertices, and that the gap between $\mathfrak{I}$ and $\mathfrak{I}_S$ opens at 14.

8. Data and code availability

The repository https://github.com/Geometric-and-Combinatorial-Structures/minimal-I-contractible-not-strong contains: the C program used in the search (`src/atascadas.c`); the SLURM scripts exactly as submitted and the source as compiled for each run, with SHA-256 checksums (`yuca/`); the complete logs of the three cluster runs, including the registry of all 2,000,000 slices of the $n = 14$ search, the list of duplicated slices, the per-task accounting (`sacct`) and the counters of the filter cascade; every graph found, in `graph6` format with English headers (`data/`); dependency-free Python implementations of exact $\mathfrak{I}_S$ membership with deletion traces and of integer homology by Smith normal form (`tools/`), independent of the C code; human-readable certificates regenerated by those tools (`certificates/`); the figures and the script producing them (`figures/`); and the LaTeX source of this paper. The script `verify.sh` re-derives every finite certificate behind Theorems 6.1, 6.4 and 7.1 and the rows $n \leq 9$ of Table 2 (rows 10 and 11 with `--full-table`) and exits with status 0 only if all checks pass. Code is released under the MIT licence and data under CC BY 4.0.

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technical support. Graph enumeration used `geng` from McKay and Piperno's *nauty/Traces* [19].

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Appendix A. Proofs of the reduction lemmas

We use two standard facts. (F1) If $N(v)$ has contractible clique complex then $\Delta(G) = \Delta(G - v) \cup \mathrm{cone}(\Delta(N(v)))$ and $\Delta(G - v) \hookrightarrow \Delta(G)$ is a homotopy equivalence; if $N(u, v)$ has contractible clique complex then $\Delta(G)$ is obtained from $\Delta(G - uv)$ by attaching the cone over $\Delta(N(u, v)) * \partial(uv)$ along a contractible subcomplex, and $\Delta(G - uv) \hookrightarrow \Delta(G)$ is a homotopy equivalence [4]. In particular $\mathfrak{I}_S$-deletable vertices and edges can be deleted without changing $\mathbb{Z}_2$-acyclicity, contractibility, or membership in $\mathfrak{I}$. (F2) If $H \in \mathfrak{I}_S$ and v (resp. uv) is $\mathfrak{I}_S$-deletable in G with $G - v = H$ (resp. $G - uv = H$), then $G \in \mathfrak{I}_S$: prepend the deletion to a reduction of H.

Proof of Lemma 3.2. (a) Let G be $\mathbb{Z}_2$-acyclic, not in $\mathfrak{I}_S$, on n_0 vertices. If G has an $\mathfrak{I}_S$-deletable vertex v, then $G - v$ is $\mathbb{Z}_2$-acyclic (F1) and not in $\mathfrak{I}_S$ (F2), with $n_0 - 1$ vertices, contradicting the minimality of n_0. If G has an $\mathfrak{I}_S$-deletable edge uv, then $G - uv$ is $\mathbb{Z}_2$-acyclic and not in $\mathfrak{I}_S$ with the same number of vertices and one edge fewer, and $G = (G - uv) + uv$ is an (I4) addition. Induction on the number of edges shows that G lies in the I_4-closure of a graph with no $\mathfrak{I}_S$-deletable element, which is stuck; if G has the minimum number of edges, it has no $\mathfrak{I}_S$-deletable edge and is itself stuck.

(b) Let $G \in \mathfrak{I} \setminus \mathfrak{I}_S$ on n_1 vertices. If v is $\mathfrak{I}_S$-deletable then $G - v \in \mathfrak{I}$ (transformation (I1), since $\mathfrak{I}_S \subseteq \mathfrak{I}$) and $G - v \notin \mathfrak{I}_S$ (F2), contradicting the minimality of n_1. If uv is $\mathfrak{I}_S$-deletable then $G - uv \in \mathfrak{I} \setminus \mathfrak{I}_S$ by (I3) and (F2), has one edge fewer and is K_4-free if G is. Induction on the number of edges concludes as in (a); the graph reached is $\mathbb{Z}_2$-acyclic because it is in $\mathfrak{I}$. □

Proof of Lemma 3.3. Let $G = A \cup_u B$ be stuck. Since $\Delta(G) = \Delta(A) \cup \Delta(B)$ with $\Delta(A) \cap \Delta(B) = \{u\}$ contractible, the Mayer–Vietoris sequence gives $\widetilde{H}_*(G; \mathbb{Z}_2) \cong \widetilde{H}_*(A; \mathbb{Z}_2) \oplus \widetilde{H}_*(B; \mathbb{Z}_2)$, so A and B are $\mathbb{Z}_2$-acyclic. Because u is a cut vertex, for every $w \in V(A) \setminus \{u\}$ we have $N_G(w) = N_A(w)$, and for every edge e of A we have $N_G(e) = N_A(e)$ (a common neighbour of the

ends of e lies in A). Since G is stuck, no vertex $w \neq u$ and no edge of A is $\mathfrak{I}_S$-deletable in A: A is almost stuck at u, and likewise B. If A had only two vertices, $A = uw$ and $N_G(w) = \{u\} \cong K_1 \in \mathfrak{I}_S$, contradicting that G is stuck; so $|A|, |B| \geq 3$. If no almost stuck graph has at most m vertices then $|A|, |B| \geq m + 1$ and $|G| = |A| + |B| - 1 \geq 2m + 1$, which proves the last statement. □

Proof of Lemma 3.4. Let uv be an edge in no triangle. Since G is 2-connected, uv lies in a cycle C. The 1-chain C (over $\mathbb{Z}_2$) is a cycle; the boundary of any 2-chain has coefficient 0 on uv because no triangle contains uv, so C is not a boundary and $\widetilde{H}_1(G; \mathbb{Z}_2) \neq 0$, a contradiction. Hence every edge lies in a triangle and every $N(u, v)$ is non-empty; since G is stuck, $N(u, v) \notin \mathfrak{I}_S$, so $|N(u, v)| \geq 2$. Now fix v and let $H = N(v)$, a graph with $|H| \geq 1$ and $H \notin \mathfrak{I}_S$, in which every vertex u satisfies $N_H(u) = N(u, v) \neq \emptyset$ with $|N_H(u)| \geq 2$. A graph with at most three vertices in which every vertex has at least two neighbours is $K_3 \in \mathfrak{I}_S$; hence $|H| \geq 4$, i.e. $\delta(G) \geq 4$. Finally, if $\deg v = n - 1$ then G is a cone with apex v, which is 0-dismantlable and hence in $\mathfrak{I}_S$, a contradiction; so $\Delta(G) \leq n - 2$. □

Appendix B. The 51 stuck K_4-free graphs on 14 vertices

Table B.4: The 51 stuck $\mathbb{Z}_2$-acyclic K_4-free graphs on 14 vertices. m = edges, t = triangles; $H_1 = H_1(\Delta(G); \mathbb{Z})$; "$\mathfrak{I}$" gives the I_4 edge certifying I-contractibility; k = length of the cycle carrying the boundary identification and "pattern" its orientation word (a^3 or $a\,a\,a^{-1}$), when the unfolding is a disc.

#	graph6	m	t	H_1	$\mathfrak{I}$	|Aut|	k / pattern
1	M?b@bFHNEk@~|k}U?	44	31	0	$(\{0,1\})$	1	5 / $a\,a\,a^{-1}$
2	M?B@thku?~Lylk}[?	45	32	0	$(\{0,3\})$	2	4 / $a\,a\,a^{-1}$ (+spur)
3	M?r@d`ifbiFfo}~o?	45	32	0	$(\{2,3\})$	2	6 / $a\,a\,a^{-1}$
4	M?B@cXamAu\r{]V{?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
5	M?B@cXemAuB}ze{]?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
6	M?BDA_wqpmJvj]|w?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
7	M?`@E_wDf[UnU}z{?	43	30	$\mathbb{Z}/3$	no	6	4 / a^3
8	M?`adQWT\LK^du~w?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
9	M?`adQWtVDUr[]^w?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
10	M?bB@_WmbRJZ|M~w?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
11	M?bB@_WsbX^Jlm~w?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
12	M?bB@_WsequuZZ~w?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
13	M?bB@_Wu`ZJjlm~w?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
14	M?bB@_WubXNJlm~w?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
15	M?bB@aWYTkZZ^M|w?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3

#	graph6	m	t	H_1	$\mathfrak{I}$	|Aut|	k / pattern
16	M?bB@aWZSmXZ|wNm?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
17	M?bB@aWZTkFf|wuu?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
18	M?bB@aWdbYFv|wuu?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
19	M?bB@aWdbZEv|wuu?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
20	M?bB@aWqplQzZ]|w?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
21	M?bB@aWyTkYZ^M|w?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
22	M?bB@aXdbY[{Nmuu?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
23	M?bB@aXlbYFfxwuu?	43	30	$\mathbb{Z}/3$	no	2	4 / a^3
24	M?bB@aXyTkYZxw^M?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
25	M?otU`SmVEKm]M^o?	43	30	$\mathbb{Z}/3$	no	4	4 / a^3
26	M?qa``{rfEEymitY?	43	30	$\mathbb{Z}/3$	no	4	4 / a^3
27	M?qa``{ufEUimitY?	43	30	$\mathbb{Z}/3$	no	4	4 / a^3
28	M?r@daiRbrTYd{tb?	43	30	$\mathbb{Z}/3$	no	1	4 / a^3
29	M?`Db_kVcyl]hyvw?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
30	M?`Db_khi{L\nUvw?	44	31	$\mathbb{Z}/3$	no	1	5 / a^3
31	M?`adL[mBkvLY{d{?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
32	M?`cu_whRRFZ~Wtm?	44	31	$\mathbb{Z}/3$	no	1	5 / a^3
33	M?`cv@UNDNZMtsY{?	44	31	$\mathbb{Z}/3$	no	1	5 / a^3
34	M?b@bCw`usLvvw{]?	44	31	$\mathbb{Z}/3$	no	1	5 / a^3
35	M?qa``[NNEYlme~o?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
36	M?qa``[iUEylNm~o?	44	31	$\mathbb{Z}/3$	no	1	5 / a^3
37	M?qa``[icjk^Nk~o?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
38	M?qa``[iej[ZNk~o?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
39	M?qa``kLeE^wt]i}?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
40	M?qa``kMmMIniu~o?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
41	M?qa``kifEfUtZ~o?	44	31	$\mathbb{Z}/3$	no	1	5 / a^3
42	M?qa`awkmiNdwu~o?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
43	M?qa`awwRxYj{k~o?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
44	M?qa`g{guLzwmkU]?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
45	M?qa`jSwubncvomd?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
46	M?qaejaLe[ngmh{[?	44	31	$\mathbb{Z}/3$	no	2	5 / a^3
47	M?`FSzcVBmVFs{J{?	45	32	$\mathbb{Z}/3$	no	4	6 / a^3
48	M?qa``{ws}Lxmkt[?	45	32	$\mathbb{Z}/3$	no	8	annulus
49	M?qa``{wvkVUt[}Q?	45	32	$\mathbb{Z}/3$	no	8	annulus
50	MCOfczgVB]UeM{p{?	45	32	$\mathbb{Z}/3$	no	4	6 / a^3
51	MCp`faiFf{W}][js?	47	34	$\mathbb{Z}/3$	no	2	not a disc

Appendix C. Certificates

Appendix C.1. Strong deletion sequences

The following sequences, produced by `tools/strong_contractible.py --trace` (an exact search independent of the C program), reduce $G_{14}^{a}+\{0,1\}$, $G_{14}^{b}+\{0,3\}$, $G_{14}^{c}+\{2,3\}$ and $D_{15}+\{9,12\}$ to K_1. Each line deletes a vertex v with $N(v) \in \mathfrak{I}_S$ or an edge uv with $N(u,v) \in \mathfrak{I}_S$ in the current graph; the reader may check each step with any of the tools in the repository.

```
# Deletion sequences certifying strong I-contractibility (J_S) of G + e
# for the three minimal examples (tools/strong_contractible.py, exact search).
# Vertex labels are those of the graph6 string. Each step deletes a vertex v
# with N(v) in J_S, or an edge uv with N(u) cap N(v) in J_S, in the current graph.

M?b@bFHNEk@~|k}U? +{0,1}  n=14 m=45  in J_S
   1. delete edge 0-5
   2. delete edge 0-10
   3. delete edge 0-12
   4. delete edge 8-12
   5. delete edge 1-12
   6. delete vertex 5
   7. delete vertex 10
   8. delete vertex 3
   9. delete vertex 6
  10. delete vertex 2
  11. delete vertex 7
  12. delete vertex 1
  13. delete vertex 0
  14. delete vertex 4
  15. delete vertex 8
  16. delete vertex 11
  17. delete vertex 12
  18. delete vertex 9
  -> K_1 after 18 deletions

M?B@thku?~Lylk}[? +{0,3}  n=14 m=46  in J_S
   1. delete edge 0-9
   2. delete edge 0-13
   3. delete edge 9-12
   4. delete edge 3-9
   5. delete edge 3-13
   6. delete edge 7-13
   7. delete vertex 2
   8. delete vertex 6
   9. delete vertex 3
  10. delete vertex 0
  11. delete vertex 4
  12. delete vertex 7
  13. delete vertex 5
  14. delete vertex 1
  15. delete vertex 10
  16. delete vertex 11
  17. delete vertex 9
  18. delete vertex 12
  19. delete vertex 8
  -> K_1 after 19 deletions

M?r@d`ifbiFfo}~o? +{2,3}  n=14 m=46  in J_S
   1. delete edge 2-7
   2. delete edge 2-10
   3. delete edge 2-13
   4. delete edge 6-13
```

```
   5. delete edge 3-13
   6. delete vertex 7
   7. delete vertex 10
   8. delete vertex 1
   9. delete vertex 4
  10. delete vertex 0
  11. delete vertex 8
  12. delete vertex 2
  13. delete vertex 3
  14. delete vertex 6
  15. delete vertex 11
  16. delete vertex 12
  17. delete vertex 9
  18. delete vertex 5
  -> K_1 after 18 deletions

# Dunce hat (Fig. 2A of DEFH 2023) plus the edge of Fig. 2B:
N|thHsL`sHoI`RPOwLG +{9,12}  n=15 m=47  in J_S
   1. delete vertex 11
   2. delete edge 0-9
   3. delete edge 0-12
   4. delete vertex 10
   5. delete vertex 8
   6. delete vertex 4
   7. delete vertex 0
   8. delete vertex 1
   9. delete vertex 2
  10. delete vertex 3
  11. delete vertex 5
  12. delete vertex 6
  13. delete vertex 7
  14. delete vertex 9
  15. delete vertex 12
  16. delete vertex 13
  -> K_1 after 16 deletions
```

Figures C.10 and C.11 show the sequences for $G_{14}^{b}+\{0,3\}$ and $G_{14}^{c}+\{2,3\}$ step by step, as Fig. 8 does for $G_{14}^{a}+\{0,1\}$.

Appendix C.2. Routes to a cone (Proposition 7.2)

Adding the following edges in order, each with common neighbourhood in $\mathfrak{I}_S$ at the moment of addition, turns the graph into a cone with the indicated apex (verified by `tools/strong_contractible.py`):

- G_{14}^{a}: $\{0,1\}$, $\{8,13\}$, $\{0,7\}$, $\{1,4\}$, $\{4,7\}$, $\{7,9\}$, $\{6,9\}$, $\{6,13\}$, $\{11,13\}$, $\{5,13\}$, $\{12,13\}$; apex 13.
- G_{14}^{b}: $\{0,3\}$, $\{3,5\}$, $\{10,12\}$, $\{0,10\}$, $\{0,6\}$, $\{6,8\}$, $\{1,10\}$, $\{1,4\}$, $\{10,11\}$, $\{2,10\}$, $\{10,13\}$, $\{9,10\}$; apex 10.

- G_{14}^{c}: $\{0,11\}$, $\{2,3\}$, $\{2,9\}$, $\{6,11\}$, $\{8,9\}$, $\{0,8\}$, $\{3,8\}$, $\{0,1\}$, $\{0,10\}$, $\{0,2\}$, $\{0,3\}$, $\{0,6\}$; apex 0.
- D_{15}: $\{2,4\}$, $\{3,5\}$, $\{2,8\}$, $\{9,12\}$, $\{5,7\}$, $\{1,7\}$, $\{7,10\}$, $\{0,7\}$; apex 7.

Appendix C.3. Integer homology

`certificates/integer_homology.txt` lists, for the three graphs of Theorem 6.1, the 51 stuck graphs and D_{15}, the number of simplices in each dimension, the $\mathbb{Z}_2$-Betti numbers and the integer homology groups computed by Smith normal form with `tools/clique_homology.py`.

Appendix D. Details of the cluster computation

All runs used the `cpu` partition of YUCA. Table 1 gives the sizes; here we record the information needed for an audit. For $n = 12$ (wrapper `envoltorio_n12_tabla.sh`, mode *table*, $M = 4{,}000$ slices, SLURM arrays 2079049/2079051, 7×64 cores, 8 minutes of wall time) all 4,000 slices were registered exactly once. For $n = 13$ (wrapper `envoltorio_n13_atascadas.sh`, mode *stuck*, `geng -C -d4 -D11 13`, $M = 200{,}000$, arrays 2079058/2079059, 7×64 cores, 7 h 10 min) all 200,000 slices were registered, with the counters `total` = 21,011,415,992,470, `pass_cone` = 3,335,926,431,910, `pass_chi` = 480,204,455,162, `pass_neighbourhoods` = 15,693,920,490, `pass_edges` = 218,516, `found` = 1. For $n = 14$ (wrappers `envoltorio_n14_K4libre.sh` on account p029, ascending slices, and `envoltorio_n14_K4libre_p011.sh` on account p011, descending slices; mode *stuck*, `geng -k -C -d4 -D12 14` from `nauty` 2.8, $M = 2{,}000{,}000$, arrays 2079102 and 2079156, $2 \times 7 \times 64$ cores, about 33 hours) the union of the registries contains all 2,000,000 slice numbers (0 missing), 692 slices appear in both registries with identical output, the counters are `total` = 64,856,470,381,820, `pass_cone` = 48,303,369,131,820, `pass_chi` = 1,462,235,073,536, `pass_neighbourhoods` = 108,022,349,478, `pass_edges` = 13,536,597, `found` = 51 labelled graphs, which are pairwise non-isomorphic. The files `yuca/logs/n14_listas_distintas.txt.gz`, `n14_duplicadas.txt`, `n14_sacct_n14.txt` and the `resumen` files in the repository are the primary records; the sources compiled for each run are in `yuca/as_run/` with their SHA-256 checksums.

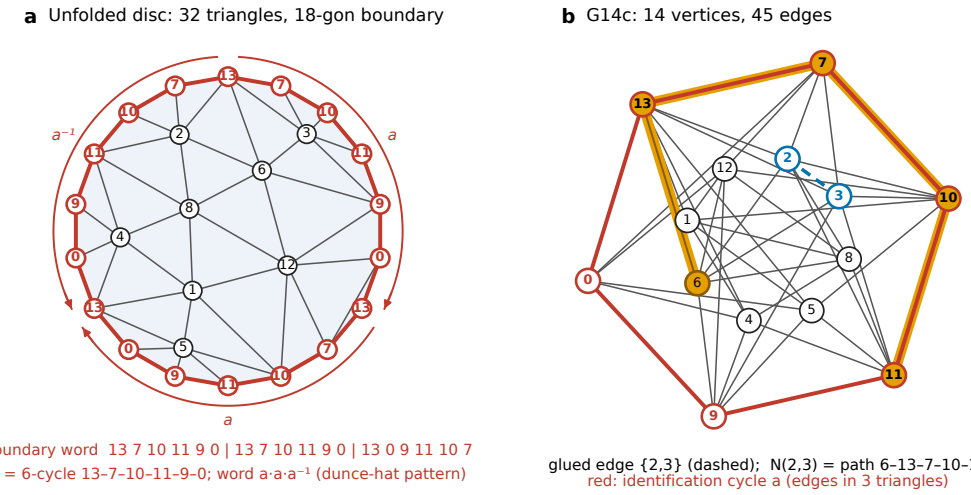


Figure 7: G_{14}^{c} (45 edges, 32 triangles, $|\mathrm{Aut}| = 2$). The boundary of the unfolded disc (26 vertices, length 18) is identified with the 6-cycle 0–13–7–10–11–9 with pattern $a\,a\,a^{-1}$. The (I4) edge is $\{2,3\}$ with $N(2,3) = 6$–13–7–10–11.

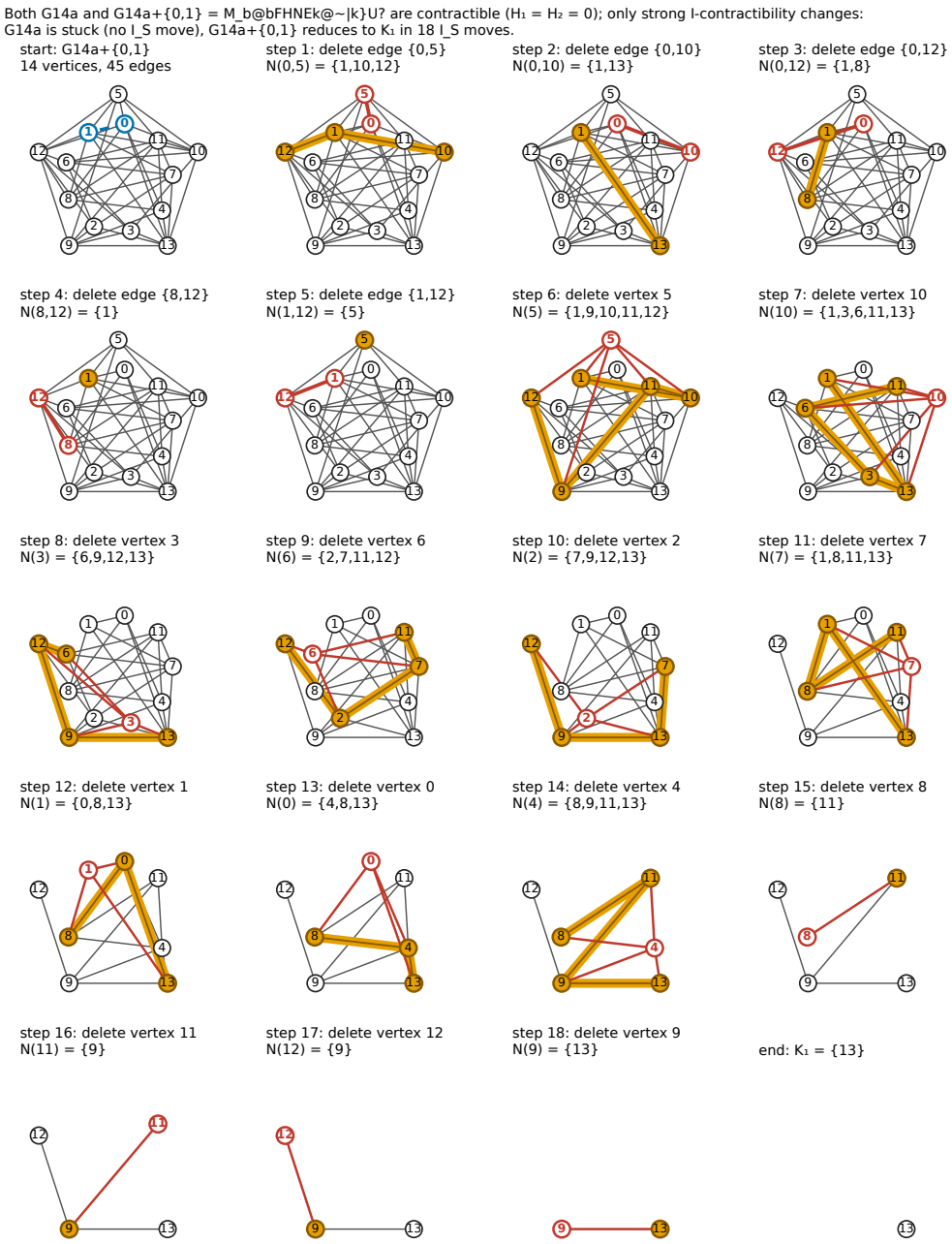


Figure 8: Certificate that $G_{14}^{a} + \{0,1\} \in \mathfrak{I}_S$: the 18 strong deletions found by Algorithm 2 (deleted element in red, its neighbourhood or common neighbourhood in amber). The sequence is listed in Appendix C and reproduced by `tools/strong_contractible.py --trace`.

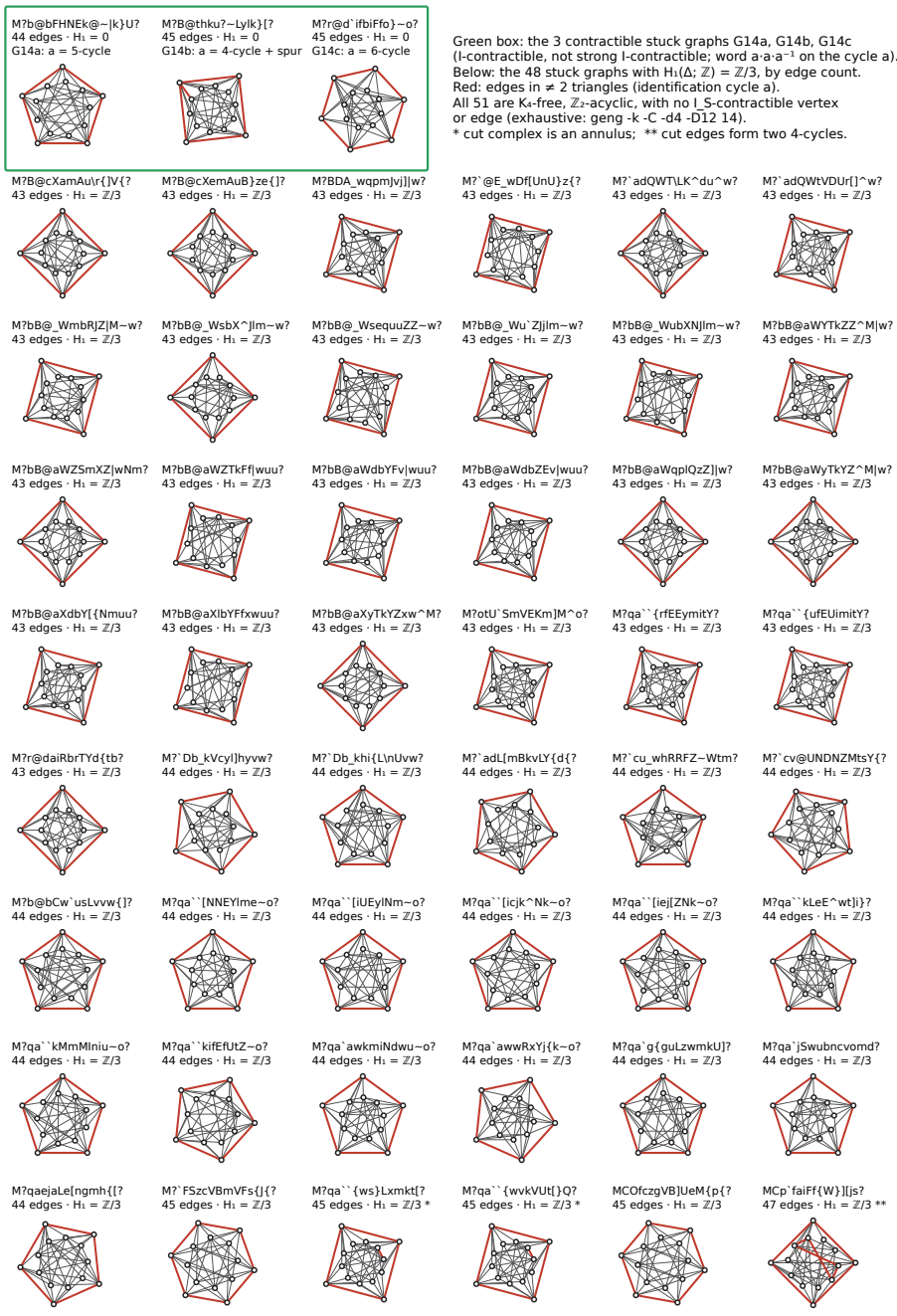


Figure 9: The 51 stuck $\mathbb{Z}_2$-acyclic K_4-free graphs on 14 vertices: the three contractible ones ($\mathfrak{I} \setminus \mathfrak{I}_S$) first, then the 48 with $H_1 \cong \mathbb{Z}/3$, ordered by number of edges.

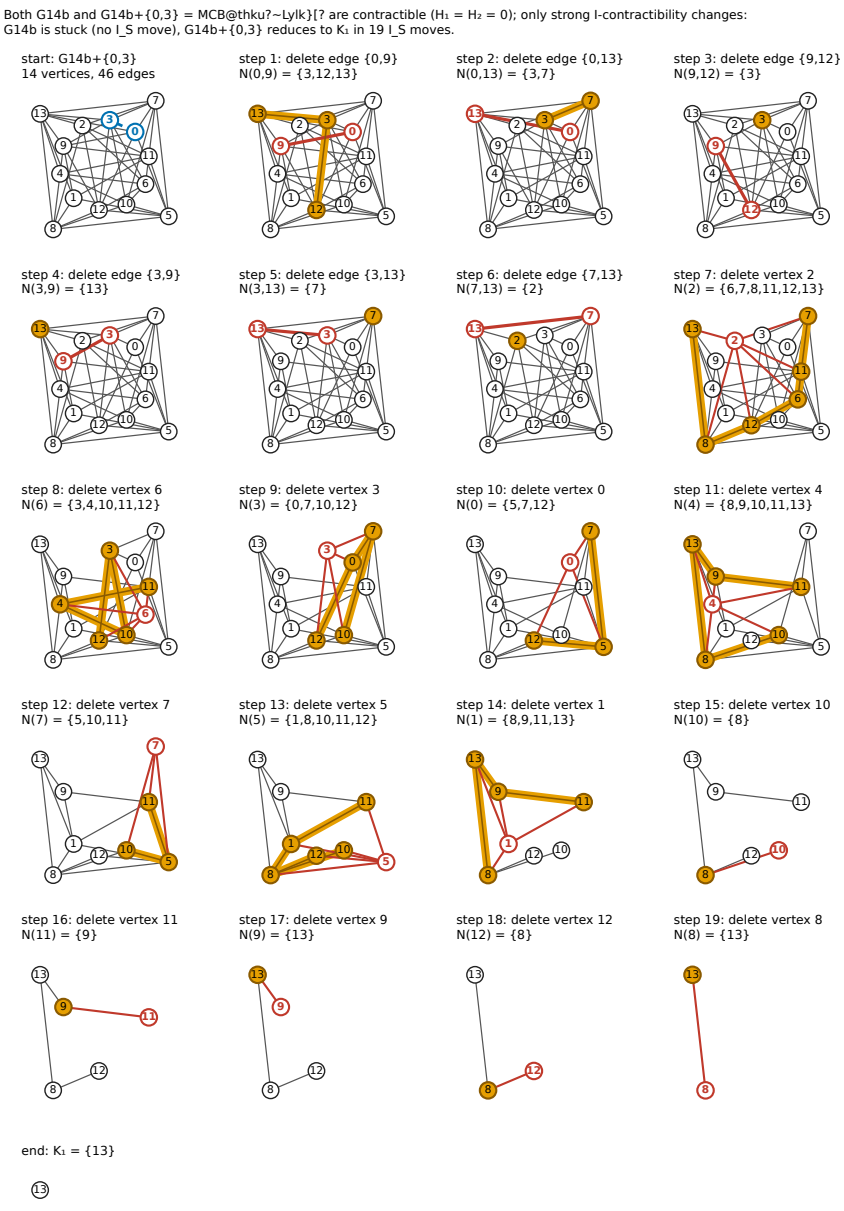


Figure C.10: Certificate that $G_{14}^{b} + \{0,3\} \in \mathfrak{I}_S$ (19 strong deletions).

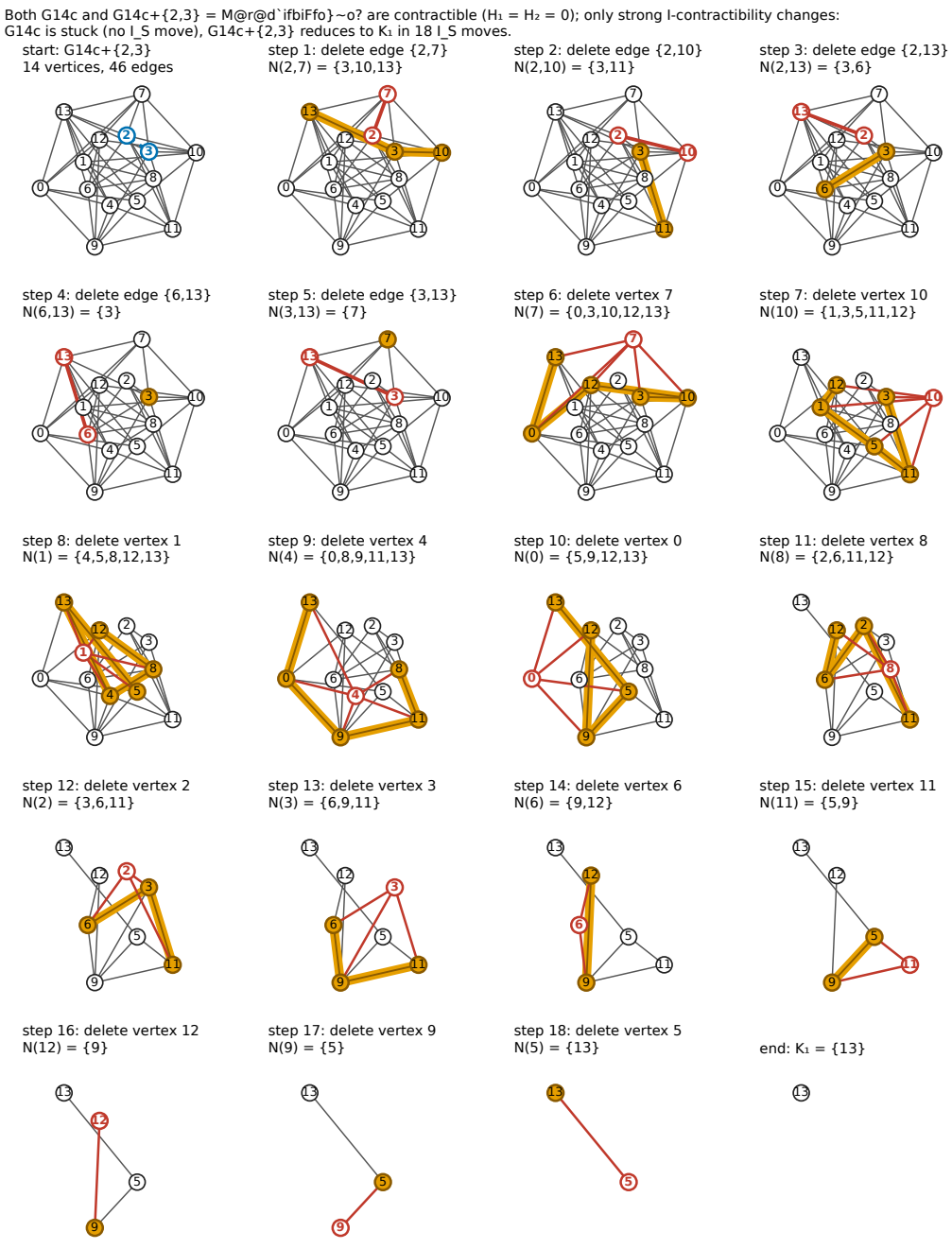


Figure C.11: Certificate that $G_{14}^c + \{2,3\} \in \mathfrak{I}_S$ (18 strong deletions).